

Face-Centered Spatial User Interfaces on Smartwatches

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ABSTRACT

Many smartwatches are now equipped with front-facing cameras that can detect users' faces and track the devices' spatial location relative to the faces in mid-air space. This space can be used as virtual placeholders for user interfaces where users can access them by moving their watch. In this paper, we present a novel face-centered spatial user interface for smartwatches that leverages the mid-air space for augmenting virtual user interfaces. With a pilot study, we first examine the suitable mid-air space that users can access during active use while wearing smartwatches. Next we conduct an online survey investigating users' attitudes towards using the space to access mid-air user interfaces. More specifically, we focus on the social acceptance of face-centered spatial user interfaces in different locations and in front of different audiences. Results indicate that participants welcomed the idea of face-centered spatial user interfaces, however, the acceptance varied based on where and in front of whom they are using the space.

CCS CONCEPTS

• **Human-centered computing** → **Smartwatch Interaction**; *Interaction space*; *Interaction techniques*.

KEYWORDS

Smartwatches, Face Input, Mid-Air Interaction, Spatial User Interface

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1 INTRODUCTION

Current advances in smartwatch design and computing power make them a potential alternative to smartphones. In comparison to smartphones, smartwatches are lighter and allow users easy access to information from their wrists without requiring much effort. However, smartwatches have many known limitations, such as limited input capabilities, finger occlusions, and fat finger issues (e.g., [6]).

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Since the smartwatch screen is very tiny, it is very hard to select a particular icon on the screen with the fingers (e.g., icons are too small to touch as the width of the finger is larger than the icon). In addition, they always require users to interact with the device with two hands - one hand wearing the watch and the other hand interacting with the on-screen contents.

To address these limitations, researchers explored ways to extend input capabilities on smartwatches by attaching external or built-in sensors. For instance, Jun et al. [10] explored one-handed gesture-based interaction by attaching proximity sensors to the smartwatches. Srinath et al. [27] investigated vision-based interaction using an external depth sensor camera to capture finger gestures on the back of the hand. On the contrary, Hongyi et al. [29] used built-in motion sensors to detect a small set of finger gestures on an unmodified smartwatch. Though many smartwatches are now equipped with a front-facing camera, to our best knowledge none of the previous work explored how to leverage this camera to extend the input capabilities on off-the-shelf smartwatches and support one-hand smartwatch usage.



Figure 1: (a) Smartwatch front-facing camera tracking the user's face during active use; (b-d) An image browser app: images from an image browser app are virtually placed into the mid-air space in front of the user, which can be accessed via moving the smartwatch around the space.

When in active use, users usually place the smartwatch in front of their faces to access screen content. In such a case, using computer vision algorithms the front-facing camera can detect users' faces and determine the spatial location of the watch relative to their face (Figure 1a). This allows users to leverage a large space in front of their faces to be augmented with user interfaces, which can be accessed by moving the smartwatch onto that space. In this paper, we present a novel face-centered spatial user interface accessible with off-the-shelf smartwatches. It uses smartwatch's front-facing cameras to detect the devices' spatial position relative to users' faces and allows access to virtual mid-air interfaces (Figure 1b-d). We first run a pilot study exploring the suitable space users can access with smartwatches. Results show that users can comfortably access a large space during active smartwatch use. We next conduct an online study exploring users' social acceptance of using face-centered spatial user interfaces in different public/private locations and in front of different audiences. We observe that participants welcome the idea of accessing face-centered spatial user interfaces

with smartwatches. However, their acceptance changes based on the locations and where and whom they are using it around.

Our contributions include: 1) A novel face-centered spatial interface for smartwatches leveraging the mid-air space in front of users; 2) An investigation of the suitable range for face-centered spatial user interface; 3) developed three applications to demonstrate the use of this novel face-centered spatial interface and 4) An exploration of the social acceptance of using face-centered spatial interface with off-the-shelf smartwatches.

2 RELATED WORK

2.1 Mid-Air Spatial Interface

Several prior works inspire the concept of face-centered spatial user interfaces. For instance, Virtual Shelves [21] is one of the earlier works exploring the concept of using mid-air spatial-interface on smartphones for triggering shortcuts. Later, researchers explored mid-air space to facilitate interactions such as placing and retrieving virtual items around the device [14], menu navigation [7], accessing buttons [13], placing virtual windows [9] or even browsing items with an e-commerce application [15]. Researchers showed that this in-air space can be a promising alternative to touch input as users are faster with mid-air space than the conventional touch input [14, 19].

Researchers commonly used external sensors to detect users' activities in mid-air space and attached them to mobile and wearable devices. For instance, IR sensors [7, 20], Omni-directional mirrors [32], depth-sensing cameras [27], and acoustic [12] have been used to track users mid-air hand and finger movements. In addition, researchers have widely been using commercial solutions, such as Vicon motion tracking system [14], OptiTrack [19], and Microsoft Kinect [16] for tracking users in-air activities. To our best knowledge, no prior work explored the use of the built-in front-facing camera on smartwatches for mid-air spatial interaction. In this paper we explored the mid-air around the face as spatial interaction space using the smartwatch's front-facing camera.

2.2 Head and Face-Based Input Spatial Interface

Researchers investigated ways to leverage head and face movement to design new interaction techniques with smart devices. For instance, Zhao et al. [34] investigated users face movement as the novel input to smartphones using on-device built-in sensors. In a similar work, Hueber et al. [18] demonstrated how head movement can be used to execute shortcuts on smartphones by tracking the head's position with the phone's front-facing cameras. Simo [5] detected users face orientation through smartphones front-facing cameras and used the head movement as input for pointing on the distant large display. Lung-Pan et al. designed iRotate [8], which can rotate iPhone and iPad screens based on face orientation which is detected by the phone's front camera. Xiaochao et al. [31] designed a face interpretation engine by calculating the pitch, roll and yaw of users faces and heads using the front-facing camera. Inspired by all these prior work, we explored face-centered spatial input on smartwatches.

2.3 Extending Smartwatch Input

Prior works have explored ways to move the smartwatch interaction space beyond the touch input. For instance, Zhu et al. [35] showed that on-device sensors (e.g., IMU) could be used to identify users finger, hand and arm movements, which can further be used for smartwatch interaction. In a similar work, Wen et al. [29] supported one-handed interaction and recognized finger gestures of the wearing hand using integrated motion sensors. Voelker et al. [28] demonstrated a calibration-free ad-hoc mobile cross-device interaction concept by estimating users' gaze directions and detecting facial features. Yang et al. [33] tracked hand postures using electromyography (EMG) sensors attached to the arm and used as smartwatch input. Han et al. [12] used an array of microphones to detect hand-made acoustics and used them to design off-device spatial interactions on smartwatches.

Researchers also explored ways to design new techniques for one-handed interaction on smartwatches. For example, Xu et al. [30] investigated one-handed smartwatch interaction using built-in accelerometer and gyroscope to identify hand and finger gestures. Gabriel et al. [22] facilitated one-handed binary object selection on smartwatches using the thumb motions in response to on-screen stimuli through magnetic sensing. Gong et al. [10] designed Wrist-Whirl to explore wrist gestures for one-handed smartwatch interaction. Though many prior works explore on-device or external sensors for extending smartwatch input space, to our best knowledge we are the first to leverage the front-facing camera on smartwatches for spatial interaction.

2.4 Social Acceptance of Input modality

Prior research has investigated the social acceptability of different interaction and input modalities. For instance, social acceptability was explored for gesture-based interaction [24], around the device interaction [2], and inputs on head-worn display (HWD) [3] in public and/or lab settings. Researchers commonly investigated social acceptance from the users' perspective, where participants were asked to use the system to have a first-hand experience [2]. On the contrary, prior works [23, 25, 26] also explored social acceptance by asking participants to watch images or video clips showing how a system is to be used by users. Participants were then asked to imagine using that system in different contexts (e.g., public/private) and provide their feedback and preferences on the system [25]. In a recent work, Alallah et al. [4] explored running social acceptance studies on crowdsourcing platforms, where they show that the crowdsourcing platforms could be a potential alternative to conduct self-paced social acceptance studies. Inspired by the results, we investigated the social acceptance of our newly introduced face-centered interaction with a popular crowdsourcing platform.

While going through prior research works, we found that most of the spatial interface solutions that we discussed above were focused on smartphone platforms. In addition, they used external attachments (e.g., IR sensors) or cameras (e.g., OptiTrack) to detect users movement around the device. None of the work focused on exploring face-centered spatial user interfaces that can be accessed

via smartwatches. In this paper, we leverage smartwatches front-facing cameras to access spatial user interfaces that are virtually located in front of users.

3 IMPLEMENTATION

3.1 Face Detection

We used Google's ML Kit [11] for Android to detect users' face from the smartwatches using the devices front-facing camera. ML Kit can detect faces and locate different facial features such as eyes, nose, and mouth on a detected face. In addition, ML Kit can detect eyebrows, lips, nose, and recognize facial expressions (e.g., whether a person is smiling or has their eyes closed).

We used LEMFO LEM14 [1] smartwatch for exploring our face-centered spatial user interfaces. It is a 4G smartwatch with one front-facing camera and one side-facing camera. We only used the front-facing camera, which is a 5.0MP camera. In addition, the watch comes with 1.6 inches round display of resolution 400×400. It has 4GB RAM and 64GB ROM and comes with Android 10.

ML Kit's face detection can be used on the smartwatch and is fast enough to be used in real-time. For instance, Salinda Hettiarachchi [17] showed that Google's ML Kit is a suitable solution for detecting faces among other face detection methods, as it could detect faces with a minimum delay (68ms) and a high accuracy. Consequently, we used their Face Detection API to detect the position and orientation of a smartwatch from the face. Note that as we were interested only in face movement and orientation, we did not use the API's face contours and facial expression features.

3.2 Input Capabilities

We designed an Android app for the smartwatch to explore the input capabilities. We leverage Google's ML Kit which detected the following relative orientations and distance of the detected face to the smartwatch camera: (a) *Yaw*: the watch's movements to the left or the right of a detected face, (b) *Pitch*: the downward or upward movement of the watch relative to the face, (c) *Roll*: tilting the watch to the right or left and (d) *Distance*: distance (in cm) between the face and the smartwatch camera.

We found that ML Kit can reliably track yaw angle between -45° and $+45^\circ$, a pitch angle between -35° and $+35^\circ$ and a distance between 5cm and 80cm (0° is located right in front of the user's face). Though the device can detect a large space in front of users, users might not feel comfortable accessing the entire space in active use while wearing the smartwatch. Consequently, we run a pilot study to explore the acceptable face-centered spatial space in front of users that can be used for placing user interfaces and be accessed with off-the-shelf smartwatches.

4 ACCEPTABLE FACE-CENTERED SPATIAL SPACE

4.1 Pilot Study

We conducted a pilot study to determine the suitable face-centered spatial space (i.e., mid-air space in front of users accessible via the smartwatch) where users can comfortably move their smartwatch to access on-screen content while wearing it on their arms. More

specifically, we were interested in determining the minimum and maximum *Yaw* and *Pitch* values to define a region which can be accessed comfortably with smartwatches. To find the region, we first designed a map application where a user can move the smartwatch in yaw and pitch directions (i.e., up, down, left and right) to pan a map. We used Google map and ML Kit APIs for Android to design the app. With the app, a user can navigate the map to the left/right or up/down directions by moving the watch to the respective locations, relative to their face. In addition, a text label was placed on the screen to show the *Yaw* and *Pitch* values.

4.2 Participants and Procedure

We recruited 6 participants (3 male and 3 female, mean age 23.8 years, all right-handed) from the local university for this study. We first asked participants to wear the smartwatch on their left wrist. We then started the map application and explained how the smartwatch leverages the front-facing camera to track face and navigate the map. After understanding the concept, they placed their hand in the natural position as they use the watch regularly and at a comfortable viewing distance. Participants were then instructed to move their hand within their comfortable range in the circular hemisphere in front of the face to navigate the map. Once they found the comfortable spatial region (i.e., minimum and maximum yaw and pitch values) to browse the map, we asked them to report the yaw and pitch values to one of the authors running the study.

4.3 Results

For each participant, We recorded the minimum and maximum yaw and pitch values representing the comfortable hand movement space along the horizontal and the vertical directions, respectively. We reported the mean of the minimum and maximum yaw and pitch values here. Results showed that users preferred to move the watch 27° to the left, 28° to the right. In addition, we found that they were comfortable moving the watch 4° to the up and 27° to the down direction. Note that the smartwatch app could capture more angular distances in the upward direction. However, participants expressed that they are not comfortable moving the hand in the upward direction, which triggers arm fatigue.

5 APP DESIGN

The pilot study showed that users could access a considerably large face-centered spatial space with smartwatches. Therefore, the space can potentially be explored with design factors such as the number of spatial user interfaces (that can be accommodated there), ways to switch between interfaces, mode switching mechanisms and other related factors. This requires running in-lab studies with the smartwatch, which was not possible due to the recent restrictions related to the omicron variant. In addition, it was crucial to explore users acceptance of such a novel face-centered user interfaces before exploring design factors. Therefore, we decided to run an online study exploring users acceptance of face-centered spatial user interfaces - investigating users attitudes regarding where and in front of whom they would use interfaces. To demonstrate the concepts to users, we designed a set of applications with face-centered spatial user interfaces. Later, we recorded videos showing an author using

the applications and collected users' attitudes on using them in public/private spaces and different audiences.

We designed three Android applications: map navigation, image browser and message history. These applications show how to use face-centered user interfaces in a self-contained smartwatch. We designed the app using Android Studio, built with Android SDK 30. The apps were designed to work with any Android device with OS between 5.0 and 11.

5.1 Image Browser

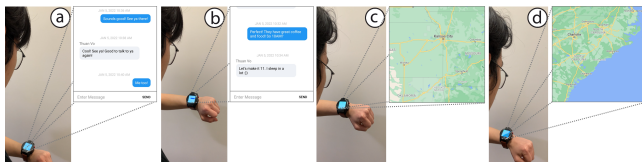


Figure 2: (a-b): Message History: a user can see message history in a messenger app by moving the watch up and down; (c-d) Map Navigation: A user can browse a map on the smartwatch by moving the hand wearing the watch.

We developed an image browsing app for the smartwatch that uses the device movements along horizontal and vertical directions to interact with the app. It leverages the mid-air space in front of users' face to offload the images into a 5×2 grid. We placed the images both horizontally and vertically into the suitable spatial space found in the pilot study. Figure 1 demonstrates a user browsing the images by moving the watch along horizontal and vertical directions.

5.2 Message History

We developed another interactive application that leverages face-centered spatial space to scroll through the message history without requiring users to touch the screen. More specifically, a user can move the watch vertically up or down between 0° (straight in front of the face) and 30° degree to scroll through the messages. Figure 2a-b presents a person browsing message history with the app.

5.3 Map Navigation

We used the same map navigation app that we used in the pilot study (Figure 2c-d).

6 SOCIAL ACCEPTANCE STUDY

Face-centered UIs require users to move their hand that is wearing the watch to left-right and up-down directions. This can generally draw the attention of co-located people in both public and private spaces. It can make users feel uncomfortable and spatial UIs less socially acceptable. However, there are specific situations (e.g., when hands are busy holding items) where this input technique can be of great benefit. Therefore, we investigated the social acceptance of this new concept of face-centered UIs by conducting an online study.

6.1 Online survey

We observed that prior works examining social acceptance mainly used two approaches: (1) providing participants with a first-hand experience using new systems/techniques and (2) showing participants a set of videos or images explaining the systems/techniques and asking them to imagine using the systems/techniques in particular contexts (e.g., public places). We could not run the study in person due to the COVID-19 pandemic restrictions. Therefore, we chose to use the second approach for our study.

We created an online survey with Qualtrics to collect feedback from participants. The survey is divided into four sections: (1) the first section presents the detailed information about the study, (2) the second section asks for basic demographic information (e.g., age, gender, if any prior experience with smartwatches), (3) the third section contains three videos showing a co-author using three applications (map navigation, image browsing, and message history) with face-centered UI, and a set of questions asking for participants' feedback on using the UIs in front of different audiences and at different locations, and (4) the last section asks participants to provide subjective feedback on the concept - rather than individual videos. The applications themselves may have different levels of social acceptance. Therefore, we clearly instruct them to focus on the idea of the face-centered user interfaces rather than the applications shown in the videos. The questionnaire contained 5-point Likert scale questions to collect participants' preferences and one open-ended question to collect descriptive subjective responses. The participants were asked to provide feedback on their comfort level - imagining themselves using face-centered UIs in front of six different audiences (i.e., colleagues, family, strangers, friends, partner, and alone) and at six different locations (i.e., home, shop, bus or train, pavement or sidewalk, pub or restaurant, and workplace). They responded on a 5-point Likert scale - Extremely comfortable, Somewhat comfortable, Neither comfortable nor uncomfortable, Somewhat uncomfortable, and Extremely uncomfortable.

6.2 Procedure and Participants

We recruited participants by posting the survey as a task on the popular crowdsourcing platform (i.e., Amazon Mechanical Turk). The survey was designed as a self-paced survey where the participants were clearly instructed to watch the video clips first and then answer the corresponding questions. They were also instructed to only focus on the mental and social aspects of using this technique while answering the questions. There are concerns of a lack of direct supervision of the participants for the online studies, thus we asked the participants to select a specific option in a question to ensure that the participant was human. We received responses from 46 participants. However, we removed incorrect/incomplete responses from 4 participants. Total 42 participants participated (20 male, mean age 34.8, SD 9.98). We compensated them with USD \$1.50 for their participation.

6.3 Result

For analyzing the data, we used non-parametric analyses and reported the median values. Figure 3 presents the result of the study.

Audience: We performed a Friedman test to analyse the audience ratings. We found a statistically significant difference in social

acceptance rating across audiences ($\chi^2(5, N = 42) = 42.56, p < .001$). To locate the differences, then we performed several Wilcoxon tests (Bonferroni: α -level 0.05 to 0.003) which showed difference between stranger and all other audiences ($p < .001$).

Location: Another Friedman test showed statistically significant difference in social acceptability rating across locations ($\chi^2(5, N = 42) = 45.88, p < .001$). Like the previous one, we again performed a series of Wilcoxon tests (Bonferroni: α -level 0.05 to 0.003) and found home differed from all other locations ($p < .001$).

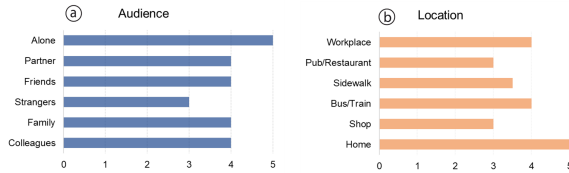


Figure 3: Medians of social acceptance rating across (a) audience and (b) location.

Subjective Feedback: We received subjective feedback where participants expressed their interest to use the UIs in private (e.g., at home) or semi-private locations (e.g., workplace) than public places. One participant commented: *Since face-centered interface is a new advancement in smartwatches, so I would feel a little awkward using it in public. I would be worried it would draw attention to myself, which I generally try to avoid while in public. I'd probably use it more at home than out in public. (Female, 37).* We also found similar feedback from participants preferring using UIs when alone or in front of more familiar audiences (e.g., family members) than strangers. For instance, we received the following comment from one participant: *I wouldn't be very comfortable using this technique with strangers, to be honest (Male, 31 year).* In addition, some participants commented that the concept is new on smartwatches; therefore, people around them would find that moving the hand around the body would be strange, unless this concept become readily available on commercial smartwatches. One participant commented: *I think it's an unfamiliar model, people around would find it a little strange at first, so I would feel uncomfortable. (Female, 24).* However, we believe that our findings align with previous results on around-device interactions [2] where researchers showed participants interest in using new techniques in front of more familiar audiences or private/semi-public locations.

7 DISCUSSION AND FUTURE WORK

Our general intuition may provide us with an idea on the suitable hand movement range (i.e., face-centered spatial space) and social acceptance of face-centered UIs. We can anticipate that the hand movement in the mid-air space is more likely to be small and participants are more likely to use this novel concept in certain public contexts. However, we need empirical data to affirm our assumptions as different factors can influence users performance and attitudes towards using the spatial interfaces. Therefore, our study results were crucial in confirming suitable hand movement ranges and users acceptance of using the novel spatial interfaces. Based on our results, we offer the following design recommendations:

Acceptable Face-Centered Spatial Space: Results from the pilot study suggest that participants were comfortable accessing spatial user interfaces and contents that are placed at a maximum 27° to the left, 28° to the right, 4° to the up and 27° to the down direction. Though users can access higher angles in all directions, we suggest designers consider the ranges mentioned above for placing UIs to reduce arm fatigue.

Considering Social Acceptance: The results indicate that participants' social acceptance of the face-centered UIs differed across audiences and locations. For instance, participants were open to using face-centered UIs in front of more familiar audiences (e.g., family members) than other audiences. Similarly, they expressed reservation on using the UIs in public locations such as in a restaurant or on a sidewalk, where large hand movements are likely to draw by-passers undesired attention and may evoke embarrassment or discomfort. Therefore, we suggest designers consider exploring social acceptance for any new technology before they get fully implemented.

Anticipated Applications: In this paper, we demonstrated three applications using the face-centered UI where the users can interact with the watch by just moving the hand wearing the watch. In this way our face-centered UI facilitates one-handed smartwatch use without any external sensors. We suggest designers to consider using face-centered interfaces while designing applications for smartwatch to support one-handed use.

We took a first step toward exploring face-centered spatial user interface from acceptable spatial space and social acceptance perspectives. Due to the recent wave of Omicron, running in-person social acceptance studies was an option available to us. In the future, we plan to run in-the-wild studies by recruiting participants from different cultures and ethnic backgrounds to get in-depth insights into these perspectives. We observed that participants who had less experience (i.e., less than two years) using smartwatches provided lower ratings than those who used smartphones for longer. Therefore, further examination is needed to explore the relationship between users' rating and their experience with technology. In addition, we aim to investigate factors associated with designing suitable face-centered spatial user interfaces for next-generation smartwatches.

8 CONCLUSION

In this paper, we explored a novel face-centered spatial user interface that leverages the mid-air space in front of users to access on-device content. With a pilot study, we first explored the suitable space that users can access with the hand wearing the smartwatch. We further investigated users' attitudes towards using the space to access mid-air user interfaces with an online study. We found that participants generally welcomed the idea of using the in-air space for accessing virtual interfaces with some reservations regarding in what public settings they would use the interfaces.

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